

Q4 — Reflection

This is the complete written response for **Q4**. The presentation uses a concise summary and links back to this evidence.

Submission baseline: v0.3.8 · execution 11901 · RUN-S2-11902-16e7090e

Supplementary demonstration evidence: execution 11958 · RUN-S2-11959-16e7090e

Published view: <https://vjra.us/prd-genie.html#q4-reflection>

Trustworthy AI comes from governed verification, not confident prose

The hard problem was preserving evidence and decisions across the delivery chain. PRD Orchestrator proved that specialized agents can accelerate extraction, gap analysis, approval, PRD generation, story decomposition, validation, and sizing. Dependable output required structured contracts, human judgment, deterministic controls, semantic evaluation, and a fail-closed release gate.

Principal lessons

- Typed intermediate outputs made missing fields, mapping errors, and invalid transitions visible.
- Ambiguity, conflict, and incomplete evidence require accountable product judgment.
- Bidirectional lineage provided stronger assurance than document-level scores alone.
- Code controls verified structure and exactness; LLM judges assessed semantic quality.
- Multiple end-to-end canaries separated repeatable behavior from a single successful run.
- Removing duplicate semantic scoring reduced cost without weakening coverage.

Accepted evidence

- Orchestration: 9 coordinated workflows.
- Authoritative inputs: 6 source documents.
- Planning hierarchy: 3 Epics · 7 Features · 11 Stories.
- Story coverage: 11/11 criteria mapped and sized.
- Release control: Agreement Gate authorized.
- Delivery: 7 validated artifacts.
- Loaded evidence: 545,467 tokens · \$1.474409.

Proposed next steps

1. Capture authoritative token usage consistently for every AI-processing stage.
2. Automate regression comparisons across documents, traceability, quality, latency, and cost.
3. Add reviewer-facing summaries for unresolved gaps, unsupported claims, and low-confidence sizes.
4. Strengthen citation and requirement reconciliation for larger source sets.
5. Introduce full-release and lower-cost routine evaluation profiles.
6. Add role-based access, retention controls, and formal security review.
7. Conduct a limited product-manager pilot while retaining the manual trigger.

Risks and controls

Risk	Potential impact	v0.3.8 control
Hallucinated requirements	Unsupported scope enters delivery	Evidence grounding, citation validation, hallucination evaluation
Loss of exact values	Dates, thresholds, or commitments change	Exact-value tests and deterministic comparison
Hidden ambiguity	Unresolved decisions appear complete	Gap analysis, confidence signals, human approval
Incomplete traceability	Teams cannot identify an item's origin	Bidirectional source-to-delivery reconciliation
Unsupported decomposition	Stories introduce unapproved detail	Story faithfulness evaluation and refinement flags
Automation bias	Polished output receives insufficient scrutiny	Explicit approval gates and visible evidence
Inconsistent output	Similar inputs produce different plans	Ground truth, regression tests, repeated canaries
Evaluation cost growth	Semantic judging exceeds generation cost	Code-first controls and duplicate-evaluator removal

Risk	Potential impact	v0.3.8 control
Sensitive-data exposure	Source content reaches unauthorized systems	Controlled credentials, limited access, security review

Connection to evaluation skills

- **Ground truth:** Independent datasets for extraction, gap analysis, approval, PRD, story breakdown, and sizing.
 - **Deterministic controls:** Schemas, counts, exact values, required sections, mappings, and release conditions.
 - **Semantic evaluation:** Faithfulness, hallucination, reasonableness, and content quality.
 - **End-to-end evidence:** n8n executions, Langfuse traces, artifacts, latency, tokens, and fully loaded cost.
 - **Error analysis:** Workflow defects, evaluator-policy issues, missing telemetry, and transient failures.
 - **Release evaluation:** A fail-closed Agreement Gate connected quality evidence to the export decision.
- Final reflection:** v0.3.8 demonstrates that AI can accelerate product documentation while accountability remains with the product team. Its value is not autonomous authorship; it is faster, more consistent, and more traceable decision support. Proposed v0.3.9 work extends that accepted foundation toward operational adoption.